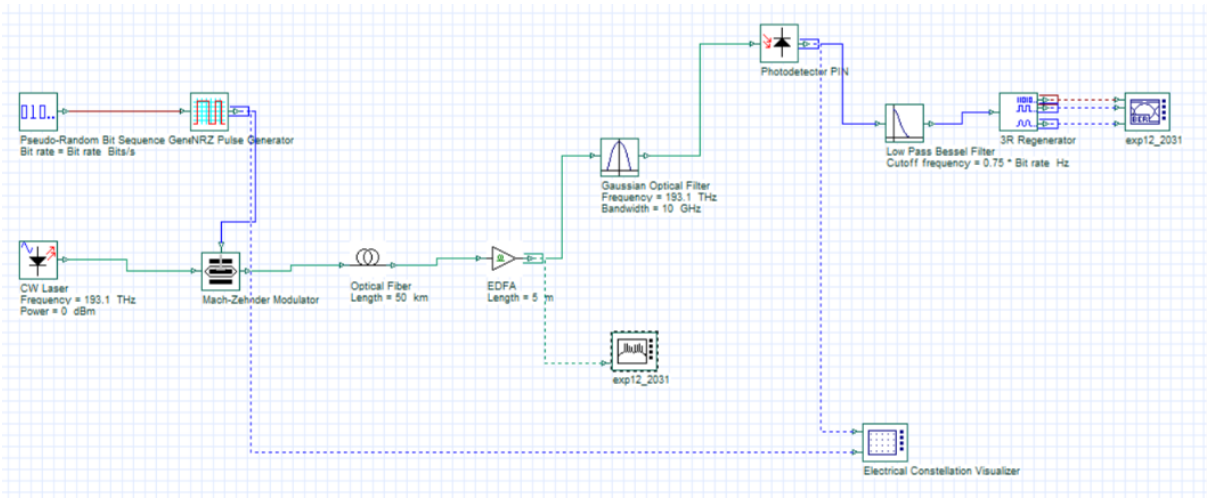
EXPERIMENT 12 : ANALYSIS OF ADVANCED OPTICAL MODULATION FORMATS (OOK, PAM-4, DPSK AND QAM) USING OPTISYSTEM

OBJECTIVES :

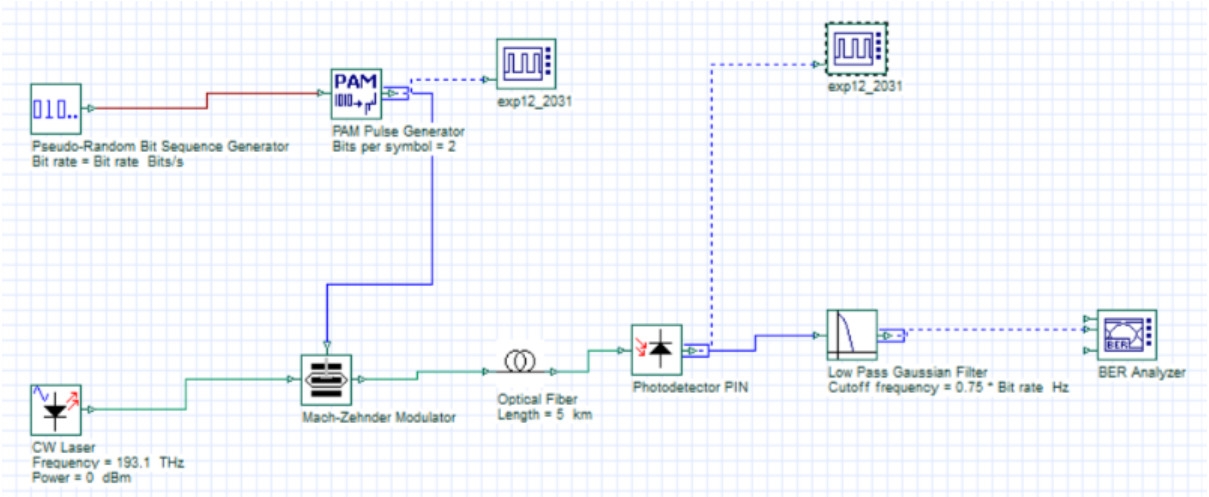
1. Design and simulate OOK, PAM-4, DPSK and QAM transmijer–receiver systems in OptiSystem.
2. Analyze the signal levels, constellation diagrams, eye diagrams and spectra of each modulation format.
3. Evaluate BER and Q-factor of an OOK link and observe the eye diagrams of the multilevel (M-ary) formats.
4. Compare the modulation formats in terms of bits per symbol, spectral efficiency and noise tolerance.

SOFTWARE USED : Optisystem software

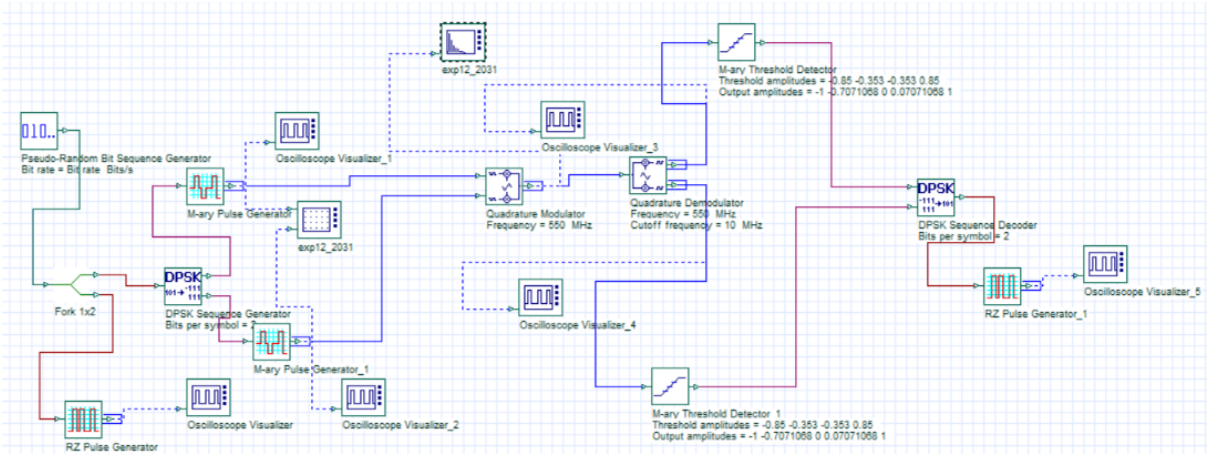
CIRCUIT DIAGRAM:



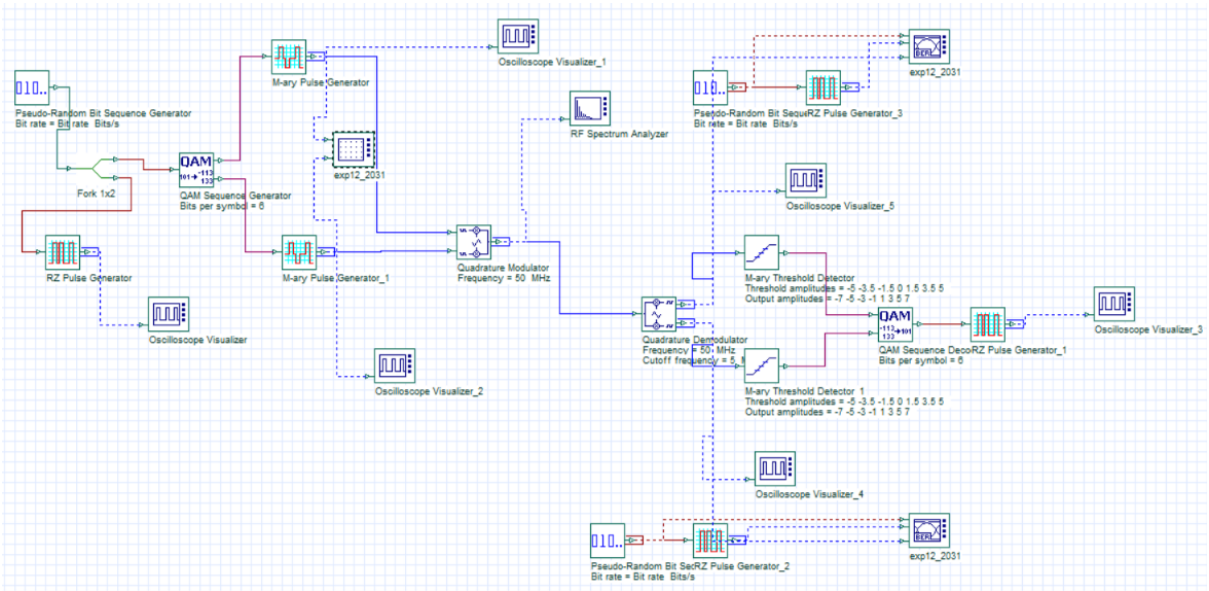
OOK transmitter and receiver



PAM-4 transmitter and receiver

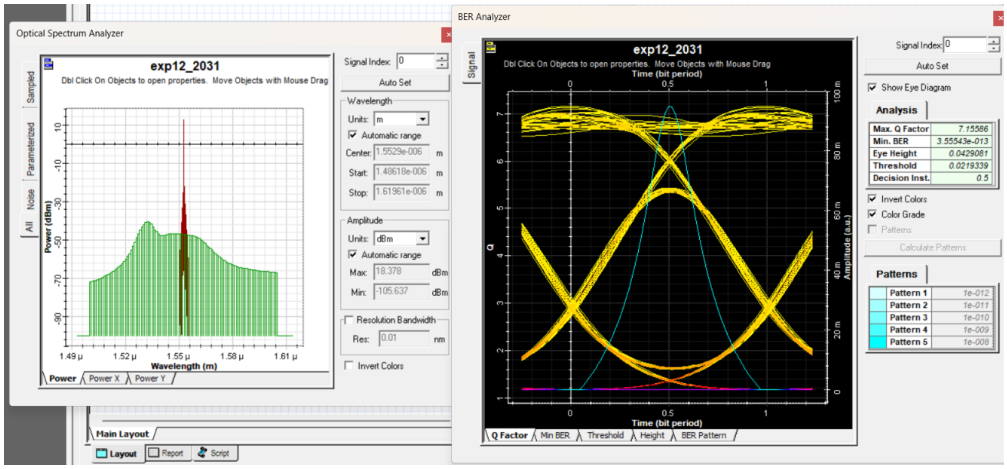


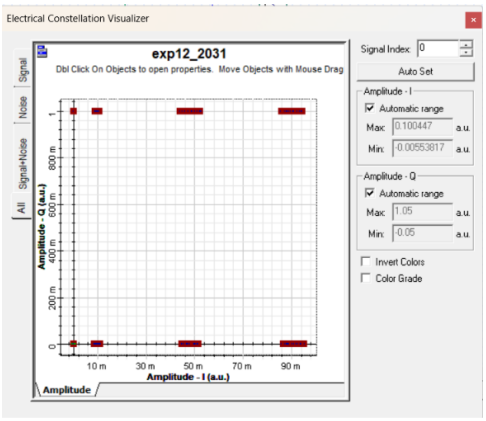
8-DPSK transmitter and receiver



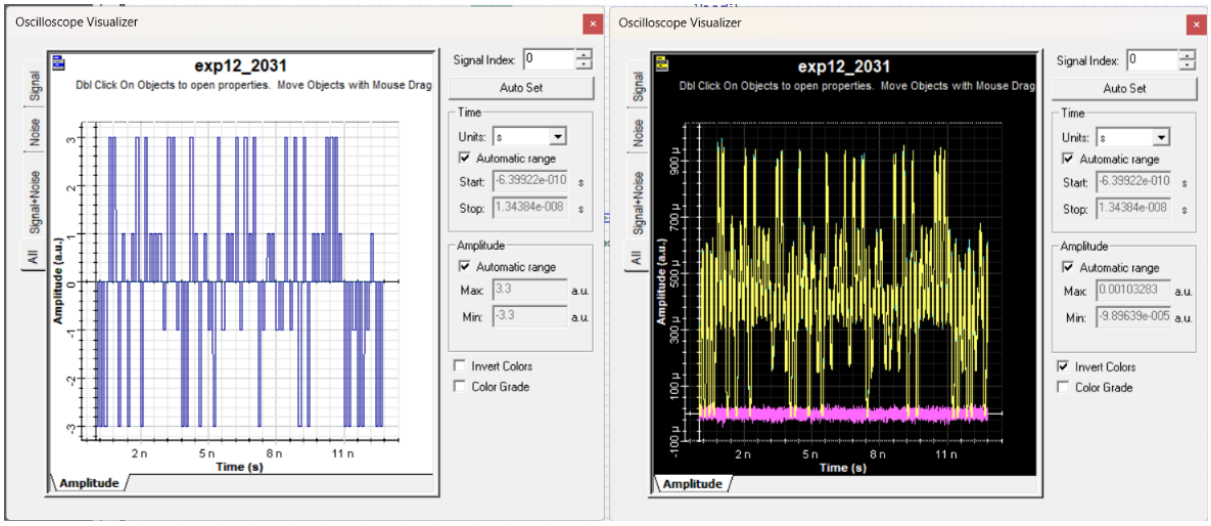
64-QAM transmitter and receiver

RESULTS:

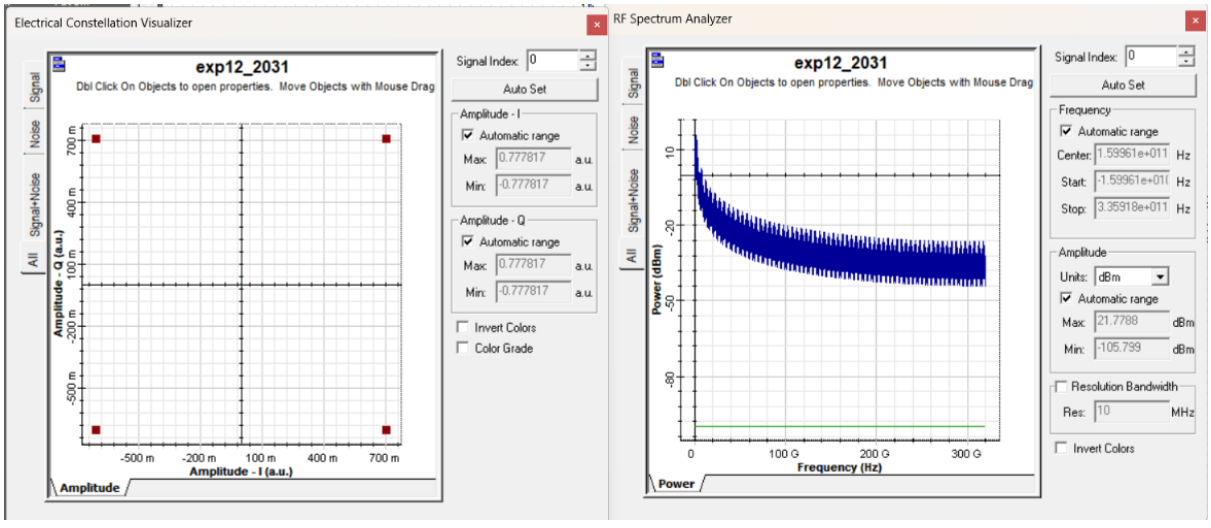




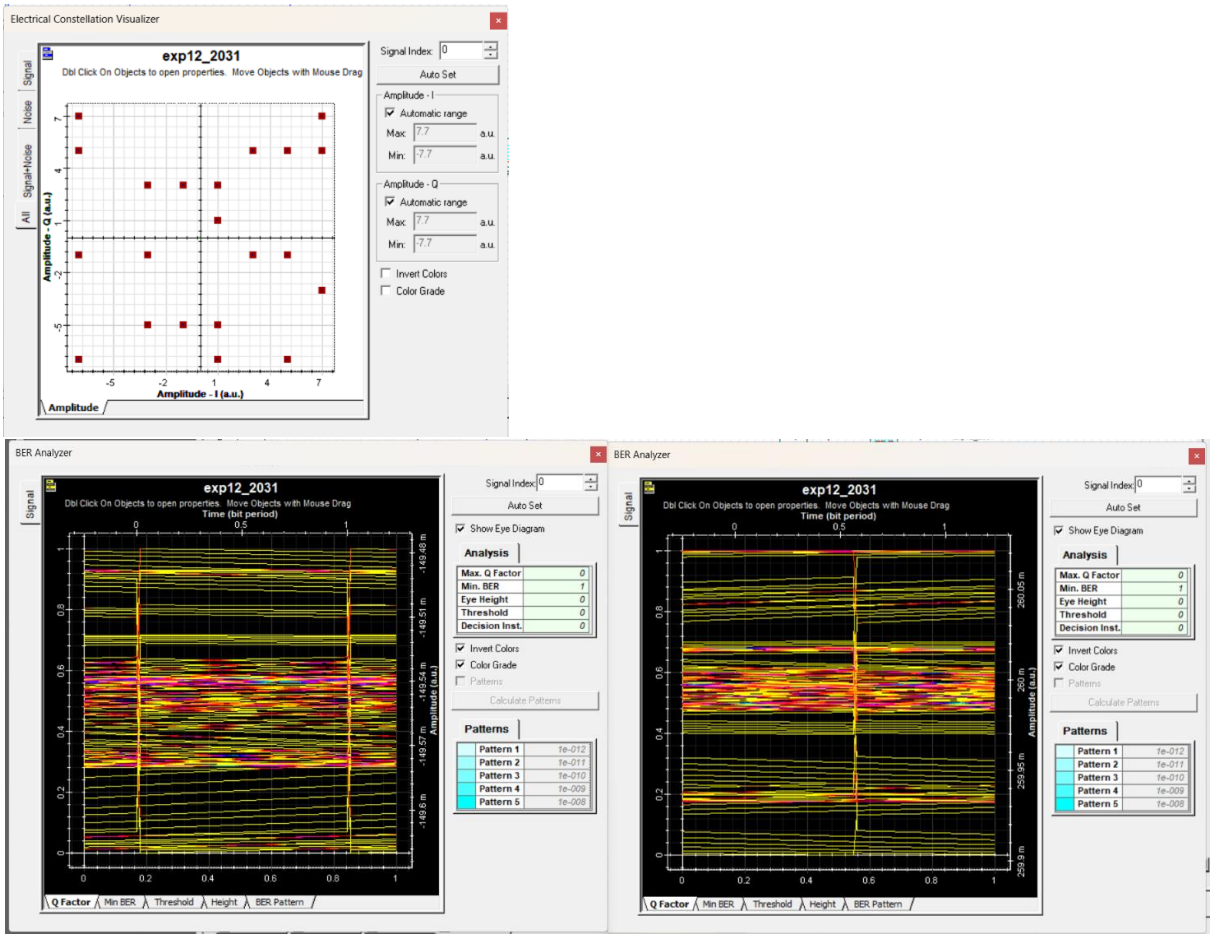
OOK transmitter and receiver



PAM-4 transmitter and receiver



8-DPSK transmitter and receiver



64-QAM transmitter and receiver

REFLECTION:

I learned to design and simulate OOK, PAM-4, 8-DPSK and 64-QAM modulation formats using Opti system and analyse their eye diagrams, and spectra. I understood how increasing bits per symbol improves spectral efficiency but reduces noise tolerance due to smaller signal separation.

RUBRICS :

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & Modulation Setup	Correct selection and connection of OptiSystem components for each format (PRBS, NRZ/PAM/DPSK/QAM generators, MachZehnder or Quadrature Modulator, fiber, PIN, filters, threshold detectors, decoders, visualizers).	All transmiLer, channel, receiver and visualizer blocks for the four formats are correctly selected, connected in sequence and configured without errors.	All components are present with minor connection or layout errors, but the systems remain functional.	Component selection is partially complete; connection or configuration errors impede signal flow.	Incorrect component selection or improper setup leading to simulation failure.	20%

Parameter Configuration & Execution	Accurate setup of bit rate, sequence length, bits per symbol, laser frequency (193.1 THz) and power, fiber length, carrier frequency and filter and threshold-detector seLings.	All parameters are accurately set for every format and all simulations are executed successfully.	Parameters are mostly accurate with minor discrepancies (e.g., sequence length or bit rate).	Basic parameters are configured, but several seLings are missing or incorrect.	Incorrect parameter seLings or failure to run the simulations.	20%
Measurement & Waveform Monitoring	Accurate recording of Q-factor, BER, eye height and threshold for OOK, and observation of M-ary signal levels, eye diagrams, constellation diagrams and spectra for PAM-4, DPSK and QAM.	Precise data extraction for all formats, neatly tabulated with clear waveforms, eye diagrams and constellation diagrams.	Data is collected with minor omissions in readings or visualizer captures.	Partial data collection with incomplete tables or missing diagrams.	Inaccurate, incomplete or missing data and measurements.	20%
Modulation Performance Analysis	Evaluation of bits per symbol, symbol rate, spectral efficiency, eye opening and constellation spacing, and their effect on noise tolerance for each format.	Comprehensive analysis correctly relating bits per symbol to spectral efficiency, eye opening and noise tolerance with sound explanations.	Clear analysis identifying basic trends among the formats.	Basic comparison without explaining the trade-off between spectral efficiency and noise tolerance.	Erroneous or missing analysis of simulation results.	20%
Report & Conclusion	Comparison of the four formats, identification of a suitable format for a given application, and structural clarity.	Report features clear diagrams, complete tables and insightful conclusions on the choice of modulation format.	Satisfactory report with relevant displays and a sound concluding summary.	Incomplete report with missing graphs, readings or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

CONCLUSION :

The simulation in OptiSystem provides a clear understanding of how OOK, PAM-4, DPSK and QAM carry data using different numbers of bits per symbol. OOK is simple and robust, while PAM-4, DPSK and QAM increase spectral efficiency at the cost of smaller eye openings, closer constellation points and lower noise tolerance. This analysis helps in selecting a suitable modulation format for different optical communication applications.